Forecasting the yield of Chinese corporate bonds

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Abstract: In this paper we focus on predicting the yield that is the centrepiece of bond markets. The dynamic Nelson-Siegel model is used to predict the yield of the Chinese corporate bonds with a class of AA, AA+ and AAA ratings. Our empirical results show that this model not only provides good in-sample fit, but also indicates the long-term, medium-term and short-term dynamic features of the yield curve of the corporate bonds with different credit ratings. Finally, we employ AR(1) model to forecast the three factors of the yield curve. Overall, the outcomes are very encouraging for the development of better forecasting systems for fixed income markets.

Keywords: corporate bonds; yield curve; Nelson-Siegel model; AR(1) model.

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1 Introduction

The corporate bond market as one part of Chinese bond markets experienced tremendous growth in size during the past decade and becoming increasing enormous since the market opened in 2007. The yield curve is the centrepiece of bond markets, and is also known as the term structure of interest rates. The yield curve of corporate bonds has become an important part of the mainstream literature in financial economics, providing insights and quantitative guidance for investment and risk management decisions.

Two popular theoretical models include equilibrium models (Vasicek, 1977; Cox et al., 1985; Duffee, 2002), which focuses on time-series dynamics but has been shown to forecast poorly, and no-arbitrage models (Ho and Lee, 1986; Hull and White, 1990), which focuses on accurately fitting the cross section of interest rates at any given time but neglects time-series dynamics, have been used extensively in this literature. Diebold and Li (2006) used neither the no-arbitrage approach nor the equilibrium approach, proposed instead a dynamic Nelson-Siegel model with three time-varying parameters interpreted as factors corresponding to level, slope and curvature. Christensen et al. (2011) proposed an affine Nelson-Siegel model without arbitrage, Ullah (2016) argued a non-arbitrage affine Nelson-Siegel model with macroeconomic and financial market variables.

Moreover, the input variables of the previous yield models in the literature have the same frequency. But the frequency of macroeconomic variables is not necessarily the same, such as GDP is quarterly frequency, CPI is monthly frequency. Therefore, Shang and Zheng (2018) developed the Nelson-Siegel model with mixed frequency data, which improves the forecast precision of the yield curve. Besides those classical methods in the field of econometrics and statistics, Liu et al. (2016, 2018b) proposed a quantum feature selection algorithm which has superior acceleration than the classical one. In addition, The suitability of machine learning methodologies for predictive analysis

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makes them particularly attractive in the context of financial asset returns predictability, such as deep learning (Tu et al., 2018; Xue et al., 2019), multi-agent recommender system (Bedi et al., 2018), logistic regression (Guo and Wei, 2019), SVM (Liu et al., 2018a) and neural networks (Zeng et al., 2018). Wu et al. (2018) focus on the optimal time of knowledge transfer in the area of big data.

In the empirical analysis we employ the dynamic Nelson Siegel model suggested in Diebold and Li (2006) to fit the yield curve of Chinese corporate bonds with AAA, AA+ and AA ratings, in order to provide investors and regulators with a method for predicting the yield of corporate bonds. The contribution of the paper adds to the recent literatures in at least two ways. First, a great majority of studies focus on the yield of US Treasury bonds or corporate bonds, which is considered as the most widely and applied one in bond markets. To the best of our knowledge, there are no published works that predict the yield of Chinese corporate bonds. Second, our findings show that, in the context of medium-term volatility and long-term volatility, the yield curve is relatively small. Moreover, the prediction accuracy of linear AR(1) models gets much better than the one of nonlinear AR(1) models. As such, we consider our research to be an important and timely contribution to this field. The remainder of the paper is as follows. Section 2 introduces the dynamic Nelson-Siegel model, Section 3 reports data and estimates fitting and predictive regression model, Section 4 contains concluding remarks.

2 Dynamic Nelson-Siegel model

In this section, the dynamic Nelson-Siegel model proposed by Diebold and Li (2006) is used to describe the yield of Chinese corporate bonds. Assuming that the maturity of the corporate bond is τ and the yield $y_t(\tau)$ at time t satisfies the following equation:

$$y_t(\tau) = \beta_{0t} + \beta_{1t}\left(\frac{1-e^{-\lambda_t\tau}}{\lambda_t\tau}\right) + \beta_{2t}\left(\frac{1-e^{-\lambda_t\tau}}{\lambda_t\tau} - e^{-\lambda_t\tau}\right) \quad (1)$$

where β_{0t}, β_{1t} and β_{2t} are the time-varying level, slope and curvature factors, respectively, λ_t is a parameter. Therefore, the dynamic Nelson-Siegel model described by equation (1) is not only simple in structure but also can reflect the long-term, short-term and medium-term dynamic changes of the yield to maturity. This paper uses the dynamic model to fit the yield of Chinese corporate bonds.

3 Experiments

3.1 Data

The price and yield data of corporate bond are sourced from the CSMAR database in the 2008-2016. For some maturities, Table 1 reports the issuance amounts of corporate bonds with different credit ratings.

Table 1 Issuance amounts of corporate bonds

Maturity (years)	*A+*	*AA-*	*AA*	*AA+*	*AAA*	*Total*
1	1	5	125	95	101	327
2	1	8	243	159	155	566
3	1	10	399	231	186	827
4	0	8	588	333	193	1,122
5	0	7	805	441	224	1,477
6	0	1	717	366	108	1,192
7	0	1	656	322	102	1,090
8	0	1	67	54	44	166
9	0	1	36	29	49	115
10	0	0	32	30	57	119
11	0	0	0	0	7	7
12	0	0	0	0	1	1
13	0	0	0	0	4	4
14	0	0	0	0	6	6
15	0	0	0	1	5	6
Total	1	12	1,117	598	380	

From Table 1, the numbers of corporate bond maturities from 1 to 10 years are much more than the numbers from 11 to 15 years. Therefore, in order to more accurately describe the yield of corporate bonds, we only calculate the yield to maturity for the remainder period of 1 to 10 years. On the other hand, the number of corporate bonds with A+ and AA- ratings is small. Therefore, we investigate the yield of corporate bonds with AA, AA+ and AAA ratings. Table 2 reports the summary statistics for the yield of corporate bonds.

Table 2 Summary statistics for the yield of corporate bonds

Rating	*Mean*	*SD*	*Min*	*max*
AA	7.182	1.236	0.000	10.000
AA+	6.637	1.306	0.000	10.000
AAA	5.558	1.332	0.000	10.000

As shown in Table 2, the mean of corporate bond yields with the higher rating is much smaller than that of corporate bonds with lower rating. For example, the average yield of corporate bonds AA rating is 7.182%, the average yield of corporate bonds with AA+ rating is 6.637%, and the average yield of corporate bonds AAA rating is 5.558%. Again, the standard deviation (SD) of the yield of corporate bonds is variable with different rating, which implies that the volatility of yields is not the same in different corporate bonds with different ratings.

Table 3 Summary statistics for estimation of three factors

Ratings	*Factors*	*Mean*	*SD*	*Min*	*Max*	$\hat{\rho}(1)$	$\hat{\rho}(6)$	$\hat{\rho}(12)$
AA	level	8.336	8.8112	–11.0363	48.0648	0.379	–0.277	–0.019
	slope	–0.4999	9.1345	–40.5489	16.5738	0.431	–0.246	–0.057
	curvature	–1.9783	12.6307	–52.8812	28.1852	0.294	–0.249	0.028
AA+	level	3.5664	19.3737	–37.5821	63.0200	0.401	0.249	0.154
	slope	2.9967	18.6325	–53.1313	42.5075	0.394	0.255	0.153
	curvature	3.7172	26.6008	–82.2491	59.1832	0.173	0.226	0.147
AAA	level	8.0287	6.6404	–2.0878	33.7102	0.044	–0.066	–0.073
	slope	0.1804	9.2839	–27.1521	138.4685	0.147	–0.023	–0.096
	curvature	–6.1588	12.5359	–44.1661	17.3044	0.044	–0.066	–0.073

3.2 Fitting the yield curve

In order to fit the yield of corporate bonds, the dynamic Nelson-Siegel model is used to estimate the parameters β_{0t}, β_{1t} and β_{2t}. The corresponding summary statistics of the three factors are shown in Table 3, where $\hat{\rho}(1)$, $\hat{\rho}(6)$ and $\hat{\rho}(12)$ represent the autocorrelation coefficients of three factors delayed 1, 6 and 12 months, respectively.

The third column in the Table 3 identifies the average value of three factors is different with different ratings, and the means of the levels and slope for the corporate bonds with AA and AA+ ratings are almost bigger than those of the curvature for the corporate bonds with AAA rating. The autocorrelation coefficient of the curvature factor of corporate bonds with AA rating is much smaller than the ones of the slope and the level factors, respectively, which shows that the medium-term volatility of the yield curve is relatively small. The autocorrelation coefficient of the slope factor of corporate bonds with AA+ rating is smaller than those of the curvature and level factors, respectively, which implies that the long-term volatility of the yield curve is relatively smaller. The autocorrelation coefficients of level, slope and curvature factors of corporate bonds with AAA rating are also small.

It is necessary to test whether three time series β_{0t}, β_{1t} and β_{2t} are stationary because the three factors are dynamic with times. Table 4 reports the results for stationary test of level, slope, curvature, respectively.

Table 4 Stationary test of three-factors

Rating	$\hat{\beta}_{0t}$	$\hat{\beta}_{1t}$	$\hat{\beta}_{2t}$
AA	–4.43*	–4.31*	–4.73**
AA+	–5.18**	–5.19**	–4.98**
AAA	–6.67***	–6.02***	–6.60***

Note: *, ** and *** indicate significance at the 10%, 5% and 1% levels, respectively.

Therefore, according to Table 4, unit root tests of the three-factors $\hat{\beta}_{0t}$, $\hat{\beta}_{1t}$ and $\hat{\beta}_{2t}$ for corporate bonds with AA, AA+ and AAA ratings are all significant at the 10%, 5% and 1% levels, respectively. It shows that the three-factors of dynamic Nelson-Siegel model fitting the yield of corporate bonds may be estimated with high efficiency. The unit root tests are one of necessary conditions for the following Autoregressive models.

3.3 Out-sample prediction

In above section, we argue that the parameters of the dynamic Nelson-Siegel model are dynamic, and the level, slope and curvature factors determine the yields of corporate bonds. So, to forecast the yields of corporate bonds is equivalent to estimate $\hat{\beta}_{0t}$, $\hat{\beta}_{1t}$ and $\hat{\beta}_{2t}$ factors. The autoregressive (AR) models are useful for predicting yields of corporate bonds. Thus the linear AR model will be compared with the nonlinear AR model in next subsection.

3.3.1 Linear AR(1) model

The forward 1-step prediction of the three factors of corporate bonds with AA, AA and AAA rating was expressed by the autoregressive AR(1) model for each kind of corporate bonds.

$$\hat{\beta}_{i,t+1} = a_i + b_i \hat{\beta}_{i,t} + e_{i,t} \tag{2}$$

where a_i is a constant, b_i is a coefficient of $\hat{\beta}_{i,t}$ (i = 0, 1, 2), and $\hat{\beta}_{i,t+1}$ is the forward 1-step prediction of $\hat{\beta}_{i,t}$, $e_{i,t}$ is the residual term. The estimation results are shown in Table 5.

Table 5 Estimation results of AR(1) model

Credit rating	b_0	b_1	b_2
AA	–0.5438** (0.1414)	–0.5431** (0.1420)	–0.5485** (0.1413)
AA+	0.3931** (0.1691)	0.3866** (0.1697)	0.403** (0.1665)
AAA	–0.6546** (0.1299)	–0.5162** (0.1363)	–0.5478** (0.1345)

Note: Parentheses report a standard error.
*** indicates significance at the 1% level.

Table 6 Estimation results of nonlinear AR (1) model

Credit rating	c_0	c_1	c_2	d_0	d_1	d_2
AA	0.3324**	0.3875**	0.081**	0.0413**	–0.0089**	0.0021**
	(0.2000)	(0.2161)	(0.2156)	(0.0180)	(0.0184)	(0.0031)
AA+	0.4816**	0.4096**	0.4280**	–0.0026**	0.0054**	0.0021**
	(0.1962)	(0.1684)	(0.1707)	(0.0047)	(0.0046)	(0.0031)
AAA	1.3665**	0.1488**	0.3755**	–0.0373**	–0.0043**	0.0034**
	(0.2223)	(0.1461)	(0.2438)	(0.0099)	(0.0068)	(0.0081)

Note: Parentheses report a standard error. ** indicates significance at the 5% level.

From Table 5, we can see that the parameters of the three factors of corporate bonds with AA, AA+ and AAA ratings are significant, which imply that AR(1) model is very efficiently. For corporate bonds with AA and AAA rating, the parameters –0.5438 are negative, which indicates that there is a reverse effect of the level factor using its current period values to forecast the next period values. This is similar to the slope and curvature factors, respectively. While it is the reverse of the corporate bond with AA+ rating. For example, the positive value of b_1 is 0.3866, which imply that the slope factor at the current period is positive effect the one at the next period.

3.3.2 Nonlinear AR(1) model

Based on the data of the three factors, there may be exist the nonlinear features, and the linear AR(1) model might not be best fitting the real feature of the three factor. Thus, in this subsection, we will employ the nonlinear AR(1) model to predict the level, slope and curvature factors corporate bonds with AA, AA+ and AAA ratings, respectively, which are given as follows:

$$\hat{\beta}_{i,t+1} = a_i + c_i\hat{\beta}_{i,t} + d_i\hat{\beta}_{i,t}^2 + e_{i,t} \tag{3}$$

where, $e_{i,t}$ is the residual term, for i = 0, 1, 2, $\hat{\beta}_{i,t}$ are the level, slope and curvature factors corporate bonds with AA, AA+ and AAA ratings, respectively. The parameter c_i indicates the linear function using the current value of $\hat{\beta}_{i,t}$ in order to forecast the next one of $\hat{\beta}_{i,t+1}$. And the parameter d_i indicates the corresponding nonlinear function.

Formula (3) is used to perform, and the corresponding results are reported in Table 6.

From Table 6, we can see that the parameters of the three factors of corporate bonds with AA, AA+ and AAA ratings are significant at the 5% significance level, which imply that nonlinear AR(1) model is very efficiently. The estimate values of the parameters c_0, c_1 and c_2 are all positive, which imply that there are some positive prediction functions of the three factors of corporate bonds. However, the nonlinear forecasting effects are different with the change of credit rating from AA to AAA rating because the estimate value of the parameters d_0, d_1 and d_2 have the different symbols, such as some are positive and others are negative.

3.3.3 Out-sample prediction models

In order to compare the out-sample prediction ability of the linear AR(1) model with the one of the nonlinear AR(1) model, the standard error between the predictive value and the model value is calculated by formula (4).

$$SE = \sqrt{\frac{\sum\left(\hat{\beta}_{i,t} - \beta_{i,t}^{\text{model}}\right)^2}{n(n-1)}} \tag{4}$$

where n is the number of samples, $\hat{\beta}_{i,t}$ is the fitting value of the dynamic Nelson-Siegel model, and $\beta_{i,t}^{\text{model}}$ is the predicted value of the model.

Table 7 Standard errors of AA corporate bonds

Prediction model	*Maturity*	*1 month*	*5 month*	*10 month*	*Average values*
Linear AR(1) model	1 year	1.1464	2.6092	3.7411	2.4989
	3year	0.3442	0.8243	1.2221	0.7969
	5 year	0.2803	0.7144	1.0705	0.6884
	7 year	0.3475	0.9524	1.4838	0.9279
	10 year	0.4548	1.0489	1.5890	1.0309
	mean	*0.5146*	*1.2298*	*1.1078*	*1.1886*
Nonlinear AR(1) model	1 year	1.2035	3.1656	4.8534	3.0742
	3 year	0.3499	0.7998	1.1471	0.7656
	5 year	0.2912	0.6693	0.9724	0.6443
	7 year	0.3686	0.8415	1.1900	0.8000
	10 year	0.4747	1.5188	3.0594	1.6843
	mean	*0.5376*	*1.3990*	*2.2445*	*1.3936*

We need to set different forecasting steps when comparing short-term, medium-term and long-term forecasting capabilities. The interval time used for fitting is [t_0, t_1], where t_0 represents a start time, and t_1 is a final time. For example, the range time used to forecast corporate bonds with AAA rating is from October 2010 to October 2015, where t_0 is October 2010 and t_1 is October 2015, respectively. The value of $t_1 - t_0$ is step a size with a month as a unit. The step sizes in our both AR(1) models are selected as 1, 5 and 10 months, respectively. Tables 7, 8 and 9 report standard errors of corporate bonds with AA, AA+ and AAA ratings, where average values at last column in each table are the average standard errors of 1, 5 and 10 months for a given maturity, and black numbers in a symbol

'mean' are the average standard errors of 1, 3, 5, 7 and 10 years for a given month, respectively.

Table 8 Standard errors of AA+ corporate bonds

Prediction model	*Maturity*	*1 month*	*5 month*	*10 month*	*Average values*
Linear AR(1) model	1 year	0.7583	1.0417	1.2484	1.0161
	3year	0.3955	0.6917	0.9687	0.6853
	5 year	0.2997	0.6119	0.9013	0.6043
	7 year	0.2999	0.6201	0.9228	0.6143
	10 year	0.7190	1.0253	1.2438	0.9660
	mean	*0.4945*	*0.7981*	*1.057*	*0.7832*
Nonlinear AR(1) model	1 year	0.9305	2.0673	2.8688	1.9556
	3 year	0.42973	1.0262	1.5891	1.0150
	5 year	0.3123	0.7383	1.1261	0.7256
	7 year	0.3051	0.6829	0.9531	0.6470
	10 year	0.8699	2.5990	3.9744	2.4811
	mean	*0.5695*	*1.4227*	*2.1023*	*1.3649*

Table 9 Standard errors of AAA corporate bonds

Prediction model	*Maturity*	*1 month*	*5 month*	*10 month*	*Average values*
Linear AR(1) model	1 year	1.1616	2.5515	3.7496	2.4876
	3year	0.4758	1.5369	2.7098	1.5742
	5 year	0.3166	1.0976	2.0663	1.1602
	7 year	0.5981	1.9394	3.2541	1.9305
	10 year	0.3946	1.1464	2.0565	1.1992
	mean	*0.5893*	*1.6544*	*2.7673*	*1.6703*
Nonlinear AR (1) model	1 year	1.1758	2.5753	3.7006	2.4829
	3 year	0.4731	1.6068	2.8415	1.6405
	5 year	0.3123	0.7383	1.1261	0.7256
	7 year	0.6010	1.5737	2.5946	1.5898
	10 year	0.3942	1.2051	2.1553	1.2515
	mean	*0.5913*	*1.5398*	*2.4836*	*1.5382*

By investigating standard errors in the above three tables, we can find that the prediction standard errors of both AR(1) models increase with the increase of prediction step sizes. It also shows that the standard errors of short-term predictions are smaller than those of the long-term predictions. And for a given predictive step size, the standard errors of the prediction decrease obviously with the increase of the maturities. Moreover, the prediction errors of the nonlinear AR(1) model are larger than those of the linear AR(1) model. This implies that the linear AR(1) is much more efficient than the nonlinear AR(1) model from the above standard errors.

4 Conclusions

The dynamic Nelson-Siegel model is used for fitting and predicting the yield of corporate bonds with different credit ratings, and two AR(1) models are estimated to forecast three factors including level, slope and curvature. It is found that the predictive ability of linear AR(1) model is much better than the one of nonlinear AR(1) model. Our research results provide some methods for investors to predict the yield of corporate bonds, which has sights and quantitative guidance for investment and risk management decisions. In addition, macroeconomic factors are the main factors affecting corporate bonds in China. Thus, in order to improve the prediction accuracy of the yield of corporate bonds, the dynamic Nelson-Siegel model with macroeconomic factors will be developed in the future.

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